华芯科技有限公司
HUAXIN TECHNOLOGY CO.,LTD

SPECIFICATION
FOR
LCD MODULE

Customer : ____________________

CustomerP/N ____________________

Model No. : ESHX497RAC11B

Version : 0

Date : 2023-07-29

Final Approval by Customer

LCM Machinery OK ☐	Checked By	
LCM Display OK ☐	Checked By	
LCM NG ☐ LCM OK ☐	Approved By	

Confirmed :

DESIGN	CHECK	APPROVAL

CONTENT

1. Version Record

Rev.	Date	Description
V0	2023-7-29	Preliminary Specification Release

2. General Specifications

Item	Contents	Unit
LCD Type	AMOLED	/
Diagonal Size	4.97	inch
Display Color	16.7M(RGBx8Bits)	
Viewing Direction	IPS(Full View)	O'Clock
Module Outline(WxHxD)	64.12*116.72*0.643	mm
Active Area(WxH)	61.884*110.016	mm
Number of Dots	720(RGB)*1280	/
Pixel Pitch	28.65*85.95	μm
Driver IC	Display: RM67295 Touch: ZT7548	/
Glass thickness (LTPS/encapsulation glass)	0.20/0.30	mm
Interface Type	MIPI 4 Lanes	/

3. Absolute Maximum Ratings

Parameter	Symbol	Spec			Unit	Note
		Min.	Typ.	Max.		
Analog/boost power voltage	VCI	-0.3	-	5.5	V	-
VCI I/O voltage	VCI_IF	-0.3	-	5.5	V	-
I/O voltage	VDDI	-0.3	-	5.5	V	-
VSP voltage	VSP	-0.3	-	6.6	V	-
VPP(OTP power)	VPP	-	-	8.25	V	-

4. Electrical Characteristics

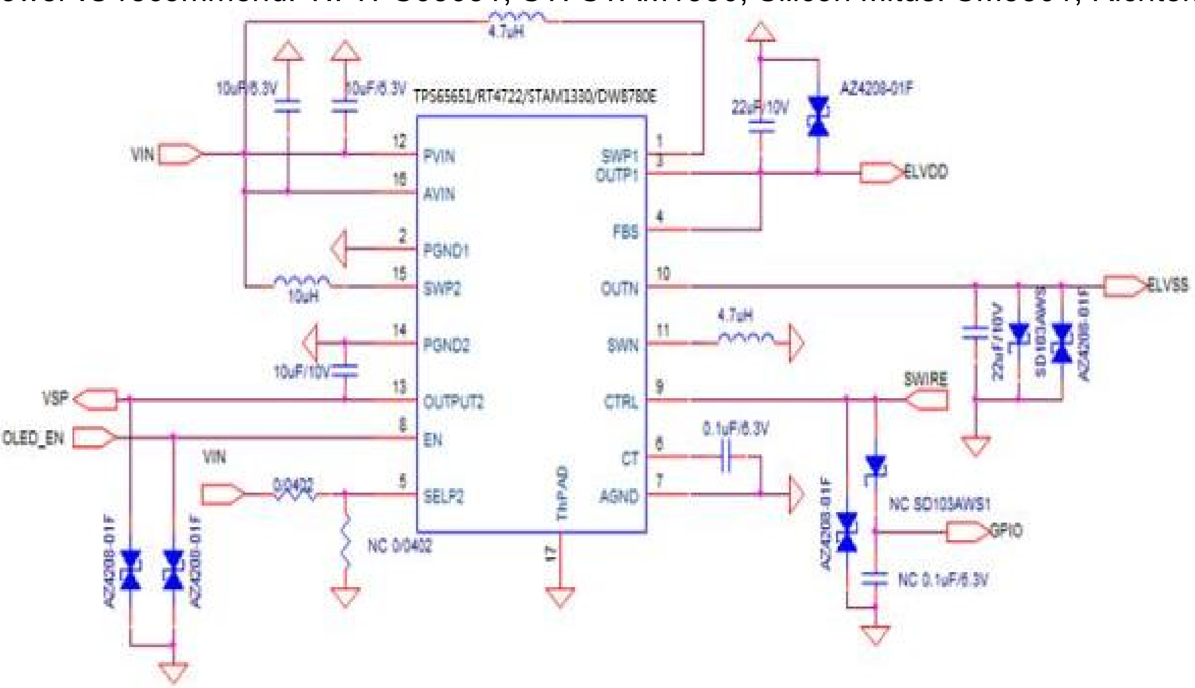
4.1 Power Characteristic

Item	Symbol	Min.	Typ.	Max.	Unit	Remark
AMOLED Power positive	ELVDD	-	4.6	-	V	
AMOLED power Negative	ELVSS	-	-2.5	-	V	Ref
Gamma Voltage	VSP	-	6.4	-	V	Ref
Digital Power supply	VDDI	1.65	1.8	3.6	V	Ref
Analog Power supply	VCI	2.5	3.3	4.8	V	Ref

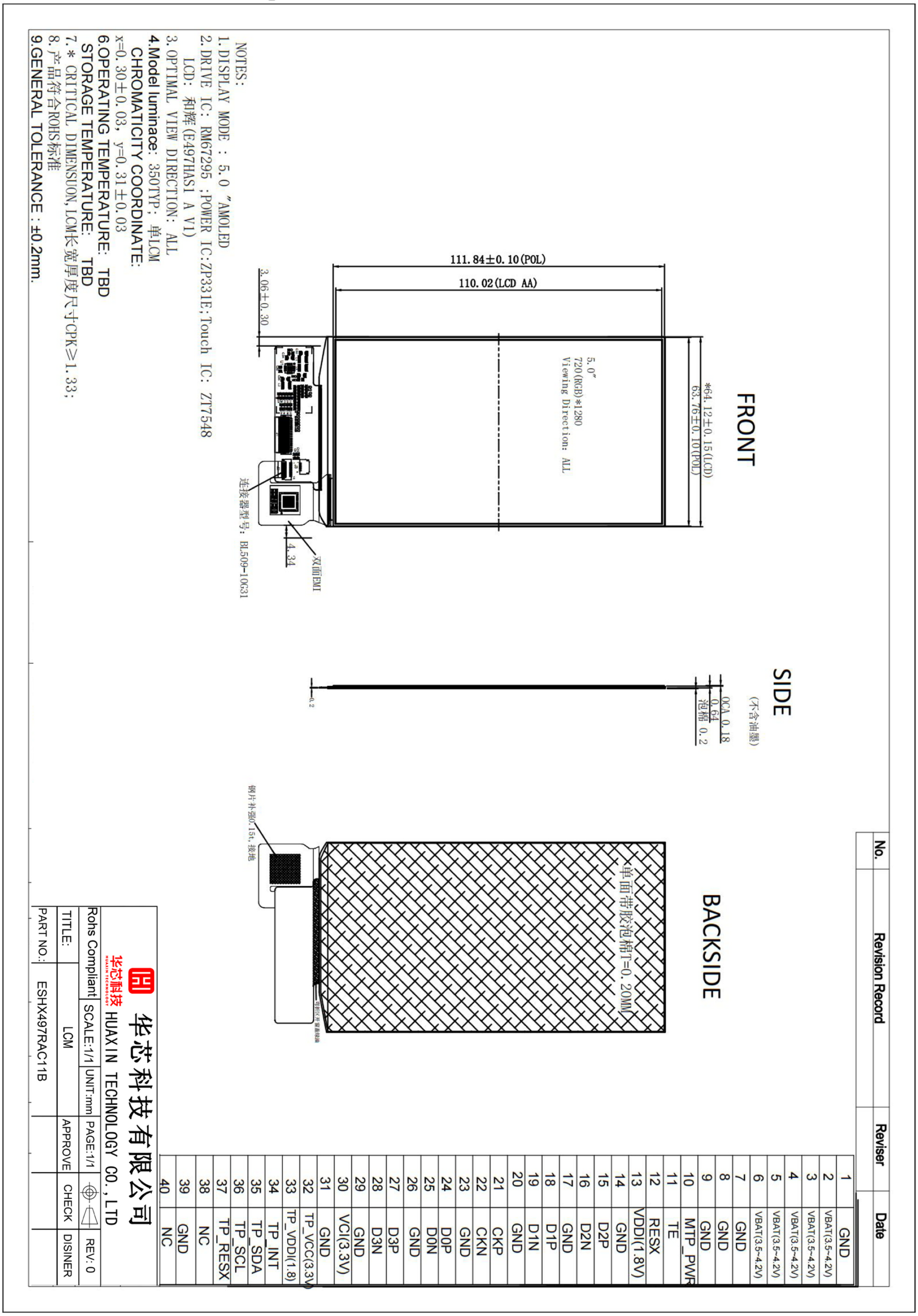
Mode	Symbol	Condition	Min.	Typ.	Max.	Unit	Remark
350 nits @Gray 255	IELVDD/E LVSS	VELVDD=4.6V VELVSS=-2.5V VCI=3.3V VDDIO=1.8V VSP=6.4V	-	160	190	mA	
	IVCI		-	1.5	2	mA	
	IVDDIO		-	15	20	mA	
	IVSP		-	10	13	mA	

4.2 Application Circuit

Power IC recommend: TI: TPS65651, ST: STAM1330, Silicon Mitus: SM3301, Richtek: RT4723



5. Dimensional Drawing



6. Interface Description

#	Pin_Name	I/O	Description
1	GND	Power	Ground
2	VBAT	Power	
3	VBAT	Power	
4	VBAT	Power	
5	VBAT	Power	
6	VBAT	Power	
7	GND	Power	Ground
8	GND	Power	Ground
9	GND	Power	Ground
10	NC	NC	
11	TE		
12	RESX	I	This signal will reset the device and must be applied to properly initialize the chip. Signal is active low.
13	VDDI	Power	Driver IC digital I/O supply
14	GND	Power	Ground
15	D2P	I	MIPI DSI data2+
16	D2N	I	MIPI DSI data2-
17	GND	Power	Ground
18	D1P	I	MIPI DSI data1+
19	D1N	I	MIPI DSI data1-
20	GND	Power	Ground
21	CKP	I	MIPI DSI clock+
23	CKN	I	MIPI DSI clock-
23	GND	Power	Ground
24	D0P	I/O	MIPI DSI data0+
25	D0N	I/O	MIPI DSI data0-
26	GND	Power	Ground
27	D3P	I	MIPI DSI data3+
28	D3N	I	MIPI DSI data3-
29	GND	Power	Ground
30	VCI	Power	Driver IC analog supply
31	GND	Power	Ground
32	TP_VCC	Power	Touch panel analog supply (connect to touch FPC pin6)
33	TP_VDDI	Power	Touch Panel I/O voltage supply (connect to touch FPC pin7)
34	TP_INT	O	Touch panel interrupt output (connect to touch FPC pin8)
35	TP_SDA	I/O	Touch panel I2C data (connect to touch FPC pin10)
36	TP_SCL	I/O	Touch panel I2C clock (connect to touch FPC pin9)
37	TP_RESX	O	Touch panel reset (connect to touch FPC pin11)
38	TP_A4	I/O	Touch panel reserved GPIO (connect to touch FPC pin12)
39	GND	Power	Ground

7. Optical Characteristics

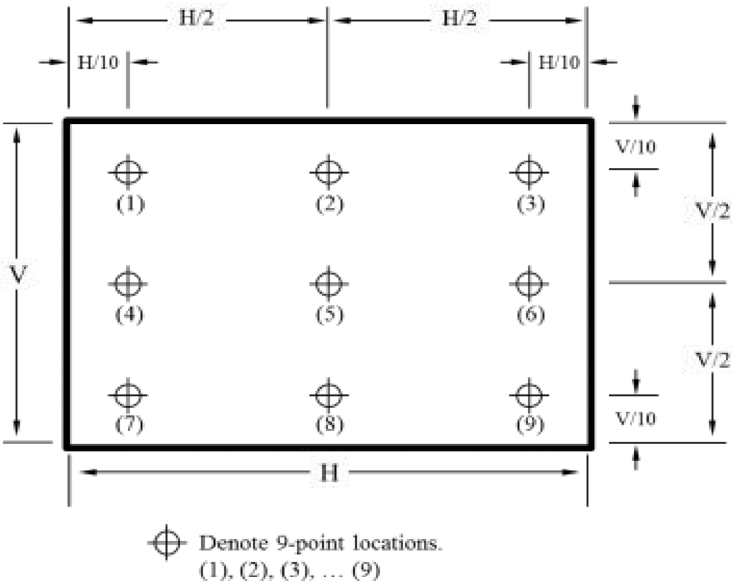
Item		Symbol	Conditions	Min.	Typ.	Max.	Unit	Remark
Brightness				315	350	385	cd/m2	Suggest MDL Brightness
Brightness Uniformity				75	85	-	%	Note 1
Contrast Ratio		CR		10000	20000			Based on CA-310 Note 2
CIE	Red	x		0.660	0.690	0.720	-	Set F
		y		0.281	0.311	0.341	-	
	Green	x		0.195	0.235	0.275	-	
		y		0.680	0.720	0.760	-	
	Blue	x		0.113	0.143	0.173	-	
		y		0.015	0.045	0.075	-	
Color Gamut			vs. NTS（CIE 1931）	90	105	-	%	
Viewing angle	Left	θL	CR≥10	75	80	-	Deg.	Note 3
	Right	θR		75	80		Deg.	Note 3
	Top	ϕT		75	80		Deg.	Note 3
	Bottom	ϕB		75	80		Deg.	Note 3
Color Shift			White @ 30 degree			6	JNCD	Note 4
Flicker						-30	dB	Note 5
Cross Talk						1.7	%	Note 6
Polarization Direction			PdF		135		Deg.	Note 7
OLED Life Time			With a Full-white image, lighting on with brightness of 350 nits for 120 hrs.	T94>=120h				
Response time						2	ms	Note 8

Note 1: Brightness Uniformity

- Environmental conditions: Temp. 25℃±3℃ , 65±20%RH, Dark Room
- Measurement equipment: CS2000 or similar equipment.
- The brightness uniformity is calculated by using following formula: Brightness uniformity = Bri.(Min.) / Bri.(Max.) × 100%

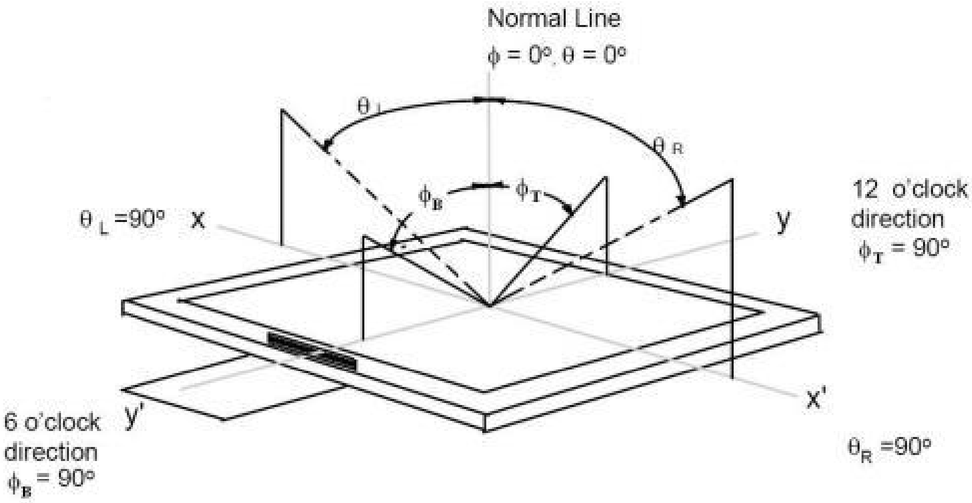
Bri.(Min.) = Minimum brightness measured in 9 measuring spots.
Bri.(Max.) = Maximum brightness measured in 9 measuring spots.

- Illustration of 9 measuring spots as follows



Note 2: Contrast Ratio
Dark Room C.R=LW/LB
LW : Full white brightness of display center P0;
LB : Full black brightness of display center P0.

Note 3: Viewing Angle
Refer to the figure below marked by θ and ϕ .



Note 4: Color Shift JNCD

- For JNCD measure, fix on white pattern, on the condition θ = 0o, φ= 0o, a color coordinate (u, 1 , v, 1) can be obtained and another color coordinate (u, 2 , v, 2) can be obtained on θL=30o.
- Delta u, v, = square root ((u, 2-u, 1)^2+(v, 2-v, 1)^2), and JNCD stands for"Just Noticeable Color Differencen . For the (u, , v,) color space 1 JNCD = 0.004 Delta u, v, , For example, color shift less than 2 JNCD means Delta u, v, <0.008.

Note 5: Flicker

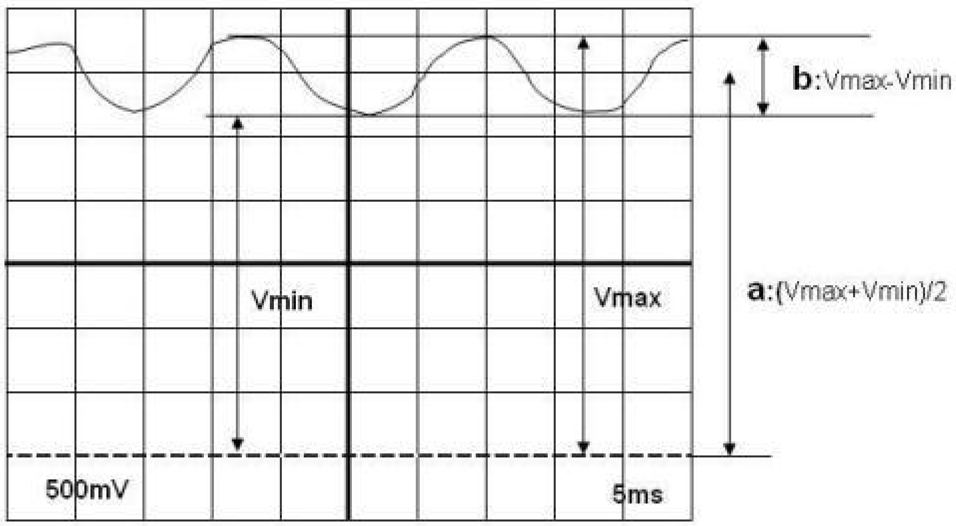
- Measurement equipment: CA-210 or similar equipment.
- Measuring temperature: Ta = 25 oC.
- Test method: JEITA method.
- Test pattern: Refer to below (Test pattern should be full-fill of display screen).

- The point should be marked is, the background of Flicker test pattern – "gray"is defined as middle grayscale. For example, RGB 24 bit"gray"is defined as below:

R7	R6	R5	R4	R3	R2	R1	R0	G7	G6	G5	G4	G3	G2	G1	G0	B7	B6	B5	B4	B3	B2	B1	B0
1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

- Frame frequency requirement before test: The display panel must be tuned to more than 60 Hz before measurement.
- If the intensity level of the display changes as Figure below, it is considered that AC component (b) overlaps on the DC component (a). With the contrast method, the ratio of AC component to DC component is defined as the flicker amount.
- AC component (b) is defined as Vmax - Vmin and DC component (a) as (Vmax + Vmin)/2, and the flicker amount is calculated by the following formula:

Flicker amount = AC component / DC component = b/a
= (Vmax - Vmin) / [(Vmax + Vmin)/2] × 100%



Note 6: Crosstalk

- Measurement equipment: CS2000 or similar equipment.
- The background of crosstalk test pattern“gray”is defined as middle gray scale. For example, RGB 24 bit“gray”is defined as below

R7	R6	R5	R4	R3	R2	R1	R0	G7	G6	G5	G4	G3	G2	G1	G0	B7	B6	B5	B4	B3	B2	B1	B0
1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

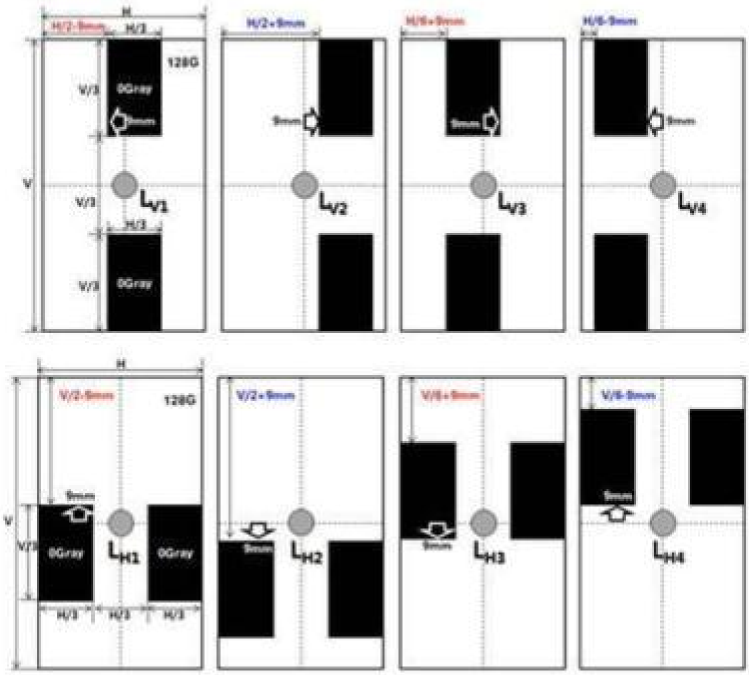
- Test pattern follows the picture below, the background is middle gray and with two black rectangle parts, each one is 1/9 of the AA size.
- Calculate the crosstalk (V) and crosstalk (H) with the formula below:

$$Crosstalk\ (V) = \max\left(\left|\frac{L_{V1} - L_{V2}}{L_{V2}}\right| \times 100, \left|\frac{L_{V3} - L_{V4}}{L_{V4}}\right| \times 100\right)$$

$$Crosstalk\ (H) = \max\left(\left|\frac{L_{H1} - L_{H2}}{L_{H2}}\right| \times 100, \left|\frac{L_{H3} - L_{H4}}{L_{H4}}\right| \times 100\right)$$

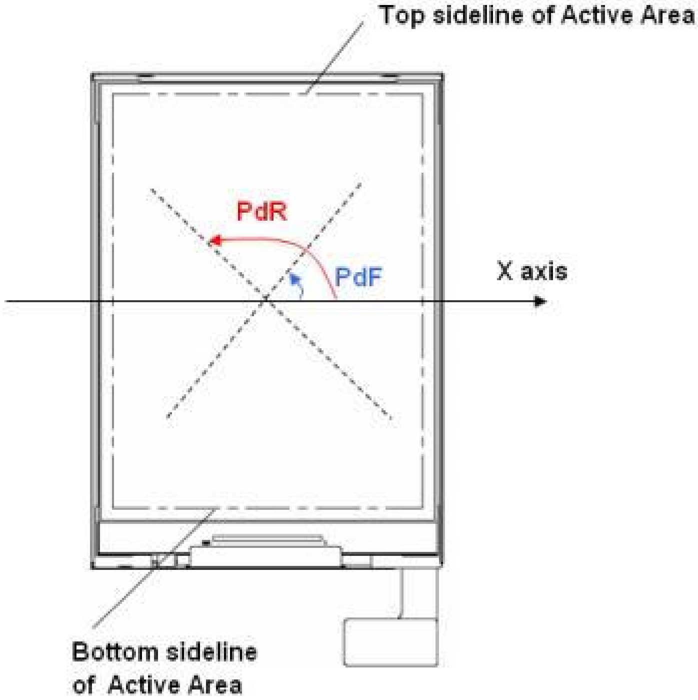
- The final crosstalk value is the maximum one between Crosstalk (V) and Crosstalk (H).

Pattern : 2/9 area is Black Box , (Background: 128gray)



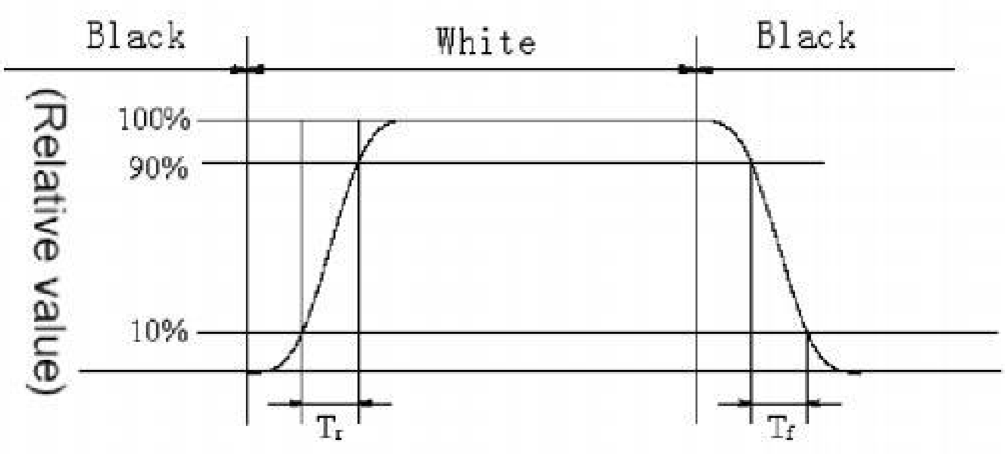
Note 7: Polarization Direction Definition

- Viewing direction is normal user viewing direction which is vertical to the display surface.
- The X axis is defined as parallel to Top and Bottom sidelines of the Active Area.
- PdF which is marked in blue arrow in the figure below is polarization degree.
- The polarization degree parameter is indicated in range of 0 deg. to 180 deg. according to above definition.



Note 8: Definition of Response Time

The output signals of photo detector are measured when the input signals are changed from“black”to“white”(voltage falling time) and from“white”to“black” (voltage rising time), respectively. The response time is defined as the time interval between 10% and 90% of the amplitudes, as shown in the figure below:



- Response time of gray to gray

Measurement equipment: CS2000 or similar equipment.

Test method: 8 grays L0 to L7 are defined, the gray level of which are 0, 36, 73, 109, 146, 182, 219, and 255. The output signals of photo detector are measured when the input signals are changed from"Lx"to"Ly", [x, y] = [0,
7]. The response time is defined as the time interval between 10% and 90%
of the amplitudes. The result of the test can be noted as below:

	L0	L1	L2	L3	L4	L5	L6	L7
L0								
L1								
L2								
L3								
L4								
L5								
L6								
L7								

8. Reliability Test Conditions

NO	Test Item	Description	Test Condition
1	High temperature storage	Endurance test applying the high storage temperature for a long time	80℃,200 H
2	Low temperature storage	Endurance test applying the low storage temperature for a long time	-30℃,200H
3	High temperature operation	Endurance test applying the electric stress under high temperature for a long time	70℃,120H
4	Low temperature operation	Endurance test applying the electric stress under low temperature for a long time	-20℃,120H
5	High temperature /humidity storage	Endurance test applying the high temperature and high humidity storage for a long time	50℃,90% R.H 200H
6	High temperature /humidity operation	Endurance test applying electric stress under high temperature and high humidity for a long time	40℃ 90% R.H 96H
7	Temperature Cycle	Endurance test applying the low and high temperature cycle -20℃ → 25℃ → 70℃ →25℃ 30min 5min 30min 5min one cycle	-20℃/60℃ 10 cycles
8	Vibration test	Endurance test applying the vibration during transportation and using	Frequency:10Hz~55Hz~10Hz Amplitude:1.5mm X,Y,Z direction for total 3hours (parking condition)
9	Fall test	Endurance test dropping the LCM from a high place	600mm height
10	Static electricity test	Endurance test applying static electric stress to terminal	Air discharge 10 times R=330Ω, C=150pF. ±8KV Remark: if malfunction can be recovered to normal state after reset or power on, it will be judged to be a good part

9. Specification of Quality Assurance

(1)Summary

The customer should check and accept the products of HuaXin within one month after reception. This standard for Quality Assurance should affirm the quality of LCD products to supply to purchaser by HuaXin. Entire process is controlled according to ISO9001

(2)Standard for quality test

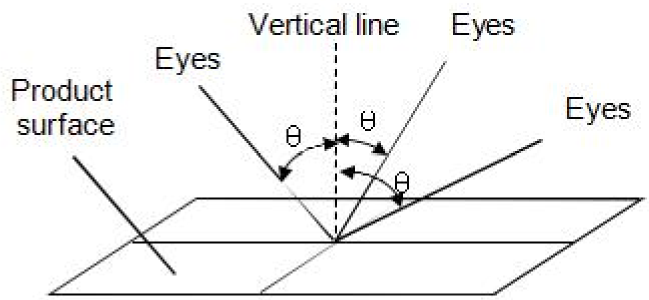
2.1) Inspection
Before delivering, the supplier should take the following tests, and affirm the quality of product.
2.2) Electro-Optical Characteristics
According to the individual specification to test the product.
2.3) Test of Appearance Characteristics:
According to the individual specification to test the product.
2.4) Test of Reliability Characteristics
According to the definition of reliability on specification for test product.
2.5) Delivery Test
Before delivering, the supplier should take the delivery test
2.6) Sampling Method: GB/T2828.1-2003, Level II
The defects classify of AQL as following
Major defect:AQL=0.65
Minor defect:AQL=1.5

(3) Nonconforming Analysis & Deal With Manners

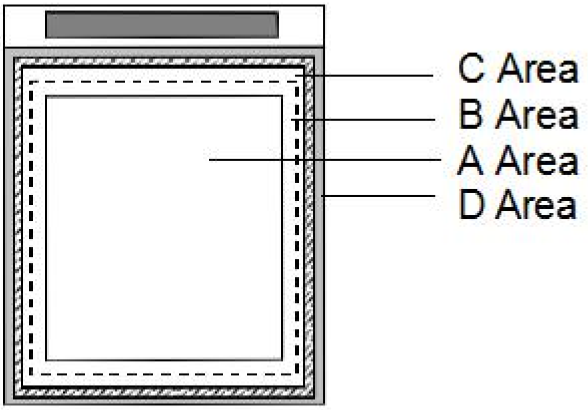
3.1) Nonconforming Analysis
Purchaser should supply the detail data of nonconforming sample and the non-suitable state.
After accepting the detail data from purchaser ,the analysis of nonconforming should be finished in two weeks.If supplier can not finish analysis on time ,must announce purchaser before two weeks.
3.2) Disposition of nonconforming
If find any supplier defect during assembly line, supplier must change the good product for every defect after recognition.
Both supplier and customer should analysis the reason and discuss the disposition of nonconforming when the reason of nonconforming is not sure
3.3) Agreement items.
Both sides should discuss together when the following problems happen:
There is any problem of standard of quality assurance ,and both sides think that must be modifier.
There is any argument item which does not record in the quality assurance. 3、Any other special problem.

(4) Standard of the Product Appearance Test

4.1) Manner of appearance test
4.2) The test must be under 20W*2 or 40W fluorescent light ,and the distance of view must be at 30±5 cm
4.3) When test the model of Transmissive product must add the reflective plate.
The test direction is base on about around 30 degree(within θ range)of vertical line.

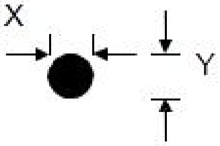


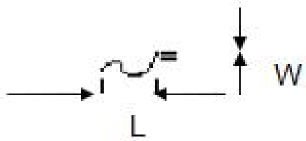
4.4) Definition of Area:
A Area: Active area
B Area: Viewing area
C Area: Out of viewing area
D Area: Seal area



(5) Basic principle:
5.1) It will accord to the AQL when the standard can not be described.
5.2) The sample of the lowest acceptable quality level must be discussed by both supplier and customer when any dispute happened.
5.3) Must add new item on time when it is necessary.

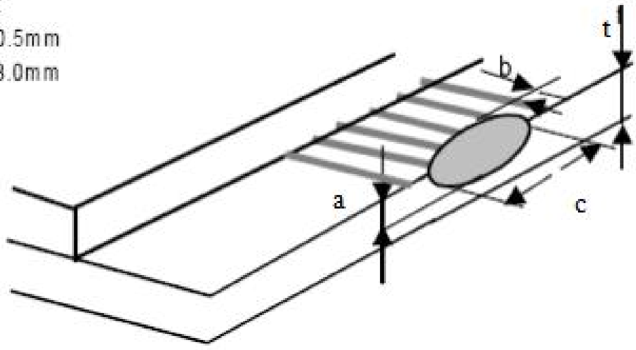
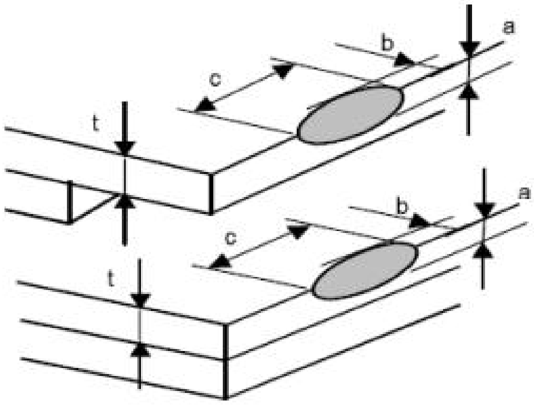
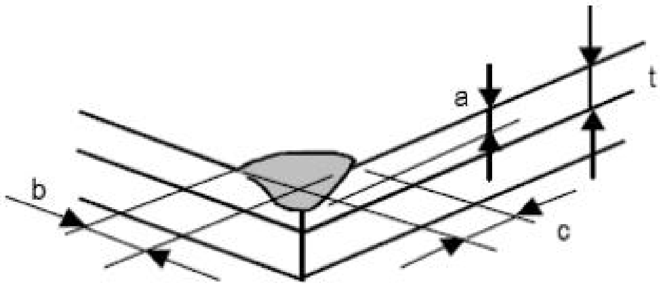
10. Inspection Criterion

<table>
<tr><th>NO</th><th>Item</th><th>Criterion</th><th>AQL</th></tr>
<tr><td>01</td><td>Electrical Testing</td><td>1.1 Missing vertical, horizontal segment, segment contrast defect.
1.2 Missing character, dot or icon.
1.3 Display malfunction.
1.4 No function or no display.
1.5 Current consumption exceeds product specifications.
1.6 LCD viewing angle defect.
1.7 Contrast defect</td><td>0.65</td></tr>
<tr><td>02</td><td>LCD black spots, white spots, color spots, contamination, scratches (display/non-display)</td><td>
2.1 Round type: As following drawing
φ=(x+y)/2


<table>
<tr><td rowspan="2">Size</td><td colspan="2">Acceptable QTY</td><td rowspan="2">Remark</td></tr>
<tr><td>A.A</td><td>V.A</td></tr>
<tr><td>φ≦0.20</td><td>Ignore</td><td>Ignore</td><td rowspan="5">No more than two spots within 5mm</td></tr>
<tr><td>0.20<φ≦0.25</td><td>2</td><td>3</td></tr>
<tr><td>0.25 ≦φ≦0.30</td><td>1</td><td>2</td></tr>
<tr><td>0.30<φ</td><td>0</td><td>0</td></tr>
<tr><td>Total</td><td>3</td><td>5</td></tr>
</table>
2.2 Line Type: (As following drawing)


<table>
<tr><td>Length</td><td>Width</td><td colspan="2">Acceptable QTY</td><td>Remark</td></tr>
<tr><td></td><td></td><td>A.A</td><td>V.A</td><td></td></tr>
<tr><td>---</td><td>W≦0.03</td><td>Ignore</td><td>Ignore</td><td></td></tr>
<tr><td>L≦2.5</td><td>0.03< W≦0.05</td><td rowspan="2">2</td><td rowspan="2">3</td><td rowspan="2">No more than two lines within 5mm</td></tr>
<tr><td>L≦1.5</td><td>0.05< W≦0.08</td></tr>
<tr><td>---</td><td>0.08< W</td><td>0</td><td>0</td><td></td></tr>
</table>
</td><td>1.5</td></tr>
</table>

<table>
<tr>
<td>03</td>
<td>Polarizer bubbles</td>
<td>If bubbles are visible, judge using black spot specifications, not easy to find, must check in specify direction.
<table>
<tr><td rowspan="2">Size</td><td colspan="2">Acceptable QTY</td></tr>
<tr><td>A.A</td><td>V.A</td></tr>
<tr><td>φ≦0.30</td><td>Ignore</td><td>Ignore</td></tr>
<tr><td>0.30<φ≦0.60</td><td>2</td><td>3</td></tr>
<tr><td></td><td>0</td><td>0</td></tr>
</table>
</td>
<td>1.5</td>
</tr>
</table>

04	Chipped glass	Symbols: a: Chip length b: Chip width c: Chip thickness t: Glass thickness 4.1 ITO electrode a<=t b<=0.5mm c<=3.0mm  4.2 General ,corner portion a<= t b<=1.0mm c<=5.0mm  *Effective width of seal area shall be more than 0.3mm. 	1.5

05	Cracked glass	The LCD with extensive crack is not acceptable.	0.65
06	Backlight elements	6.1 Illumination source flickers when lit. 6.2 Spots or scratches that appear when lit must be judged using LCD spot, lines and contamination standards. 6.3 Backlight doesn't light or color is wrong	0.65 1.5 0.65
07	Soldering	7.1 No unmelted solder paste may be present on the PCB. 7.2 No cold solder joints, missing solder connections, oxidation or icicle. 7.3 No residue or solder balls on PCB. 7.4 No short circuits in components on PCB.	1.5 1.5 1.5 0.65
08	General appearance	8.1 No oxidation, contamination, curves or, bends on interface pin (OLB) of TCP. 8.2 No cracks on interface pin(OLB) of TCP 8.3 NO contamination, solder residue or solder balls on product. 8.4 The IC on the TCP may not be damaged, circuits. 8.5 The residual rosin or tin oil of soldering (component or chip component) is not burned into brown or black color. 8.6 Sealant on top of the ITO circuit has not hardened 8.7 Pin type must match type in specification sheet. 8.8 LCD pin loose or missing pins. 8.9 Product packaging must the same as specified on packaging specification sheet. 8.10 Product dimension and structure must conform to product specification sheet.	1.5 0.65 1.5 0.65 1.5 1.5 0.65 0.65 0.65 0.65

r Use of LCD Moc

ade of glass. Do not subj

damaged and the liquid crystal substance inside it leaks out, be sure not to
e substance comes into contact with your skin or clothes, promptly wash it off

e force to the display surface or the adjoining areas since this may cause the

the display surface of the LCD module is soft and easily scratched. Handle

s contaminated, breathe on the surface and gently wipe it with a soft dry cloth.
moisten cloth with one of the following solvents:

mentioned above may damage the polarizer. Especially, do not use the

ssemble the LCD Module.
er is off, do not apply the input signals.
of the elements by static electricity, be careful to maintain an optimum

body when handling the LCD Modules.
embly, such as soldering irons, must be properly ground.
t of static electricity generated, do not conduct assembly and other work

oated with a film to protect the display surface. Be care when peeling off this
atic electricity may be generated.

modules, avoid exposure to direct sunlight or to the light of fluorescent

uld be stored under the storage temperature range. If the LCD modules will
he recommend condition is:

uld be stored in the room without acid, alkali and harmful gas.
uld be no falling and violent shocking during transportation, and also should
er, damp and sunshine.